

Lakshay Tyagi

Software Developer and Machine Learning Engineer building cutting-edge AI solutions at AWS
My work revolves around building Search, Large Language Model (LLM), Computer Vision,
Machine Learning, Retrieval-Augmented Generation, and Distributed Database Solutions

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 [Github](#)

 [Google Scholar](#)

 [LinkedIn](#)

 [Personal Website](#)

Education

New York University, Courant Institute of Mathematical Sciences

Sep 2022 - May 2024

MS in Computer Science; GPA: 4.0/4.0; Fellowship for exceptional academic performance

New York City, NY

Indian Institute of Technology, Kanpur (IITK)

Jul 2017 - May 2022

BTech, Double Major Electrical Engineering; GPA: 9.5/10.0; General Proficiency Medal

Kanpur, UP

Work Experience

Amazon Web Services (AWS) - L5 SDE - AI/ML Amazon Bedrock

Jul 2024 - PRESENT

LLM Inference & Serving

- Optimized deep-learning inference using **vLLM** and **FlashAttention**, reducing model serving latency by 10%
- Scaled embedding generation throughput by **3×** through **token-based dynamic batching** and **serverless autoscaling**
- Architected integration of **Cohere** and **Amazon Rerank** models in **Bedrock Knowledge Bases**, improving recall by 10%
- Implemented docker container based Amazon SageMaker **CI/CD pipelines**, reducing time to deployment by 80%

AI Grounding & Multi-Agent Workflows

- Orchestrated **multi-agent workflows** with automated review and verification, improving code-generation accuracy by 35%
- Developed a sandboxed **code-generation agent** using **Lambda** for answering aggregation and statistical analysis queries
- Implemented **agentic retrieval** with inferred aggregation queries, improving LLM response accuracy by 10%
- Engineered **document-freshness mechanisms** for retrieval-augmented LLM generation, improving response recency by 65%

Search & Distributed Systems

- Implemented **fuzzy metadata search**, improving response generation accuracy for metadata use-cases by 20%
- Redesigned hierarchical chunking retrieval using **NoSQL-backed** text storage, reducing retrieval latency by 200 ms
- Optimized **ElasticSearch** clusters, reducing latency by 47% and lowering annual infrastructure costs by \$150K+

Internships

NYU Langone Institute - Research Engineer - [Publication](#)

Oct 2023 - May 2024

- Developed Large Language Model (LLM) solutions for labelling medical reports with pathologies with explanations
- Pretrained and fine-tuned BERT and LLAMA models to SOTA performance with LORA on corpus of 3,400,000 medical reports

Bayer, AI Team - Data Science Intern

Jun 2023 - Sep 2023

- Orchestrated time series forecasting pipelines in Databricks for predicting sales growth, integrated with Snowflake using SQL
- Engineered features and trained XGBoost, LSTM (RNN), and graph models to reduce MAPE to 0.11 on backtests

Samsung Research Institute - Software Engineering Intern

May 2020 - Jul 2020

- Implemented Autoencoder-based deep learning models in TensorFlow for denoising low light, grainy videos
- Delivered State of the Art SSIM of 0.89 and PSNR of 30.14 for video generation using perceptual loss and sensor input

Technical Skills

Programming Languages: Python, Java, C/C++, Ruby, Scala, TypeScript; **Databases:** DynamoDB, ElasticSearch, Snowflake, Postgres

Libraries: PyTorch, Tensorflow, Keras, NumPy, Scikit-Learn, OpenCV, pandas, HuggingFace, Spring/Boot, cvxpy, Scikit-Learn, NLTK

Research Publications

- Large Language Models for Quantifying Humor and Offensiveness A Gupta*, **L Tyagi***, A Pal*, B Khurana*, A Modi: Publication in Association of Comp. Linguistics (ACL) : [ACL-ICJNLP 2021](#)
- Federated Learning Using Variable Local Training for Brain Tumor Segmentation: A. Tuladhar, **L. Tyagi**, R. Souza, ND. Forkert, [BrainLes MICCAI 2021](#)